

[POSIX-FSA] POSIX Extensions to MS-FSA

Specifies the MS-FSA extensions for supporting POSIX compliant operating systems.

Published Version

Date	Protocol Revision	Revision Class	Downloads

Previous Versions

Date	Protocol Revision	Revision Class	Downloads

1 Introduction

The SMB3 POSIX Extensions are extensions to enable POSIX compliant operating systems to better interoperate with SMB3 servers and storage appliances. This document specified the extensions to the [\[MS-FSA\]](#) specification. Popular servers such as Samba, Windows Server and others support SMB3 by default. These extensions are already implemented in multiple clients and servers.

2 Algorithm Details

2.1 Object Store Details

2.1.1 Abstract Data Model

2.1.1.1 Per Volume

SupportsPosix: A boolean that is TRUE if the volume's filesystem supports POSIX semantics.

2.1.1.6 Per Open

IsPosix: A boolean that is TRUE if this open is a POSIX open.

2.1.1.7 Per ByteRangeLock

IsPosix: A boolean that is TRUE if this lock is a POSIX lock.

2.1.3 Initialization

Volume.SupportsPosix MUST be initialized in an implementation defined manner.

2.1.4 Common Algorithms

2.1.4.2 Algorithm for Detecting If Open Files Exist Under a Directory

When processing the following condition from MS-FSA 2.1.4.2:

- If Link.File.OpenList contains an **Open** with Open.Link equal to Link

and the condition is TRUE, the object store MUST additionally check:

- If Open.IsPosix is TRUE, ignore this open.

2.1.4.10 Algorithm for Determining If a Range Access Conflicts with Byte-Range Locks

2.1.5 Higher-Layer Triggered Events

2.1.5.1 Server Requests an Open of a File

2.1.5.1.1 Creation of a New File

2.1.5.1.2 Open of an Existing File

2.1.5.4 Server Requests Closing an Open

2.1.5.5 Server Requests Querying a Directory

2.1.5.5.3 Directory Information Queries

2.1.5.5.3.1 FilePosixInformation

2.1.5.7 Server Requests a Byte-Range Lock

2.1.5.8 Server Requests an Unlock of a Byte-Range

2.1.5.11 Server Requests a Query of File Information

2.1.5.11.1 FilePosixInformation

2.1.5.12 Server Requests a Query of File System Information

2.1.5.12.1 FileFsPosixInformation

2.1.5.16 Server Requests Setting of Security Information